Assignment 2: Spellch...
General
Your Task
An Introduction to T...
Extra Insertion Exa...
Implementation Det...
Implementation Re...
Assessment
Dataset Download
Assignment Submis...
Marking Alignment ...
Reflection Submissi...
Submission Certific...
Supplementary: Sha...

Assignment 1 to account for **misspelled keys** when **querying** your dictionary.

Assignment

Overall, you will create a partial error-handling dictionary (spellchecker) using a patricia tree.

- You will be using the **same dataset** as Assignment 1.
- Users will be able to **query the patricia tree** and will get either the expected key, or the **closest** recommended key.
- You will then **write a report** to analyse the time and memory complexity of both your Assignment 1 **linked list** and your **patricia tree** implementation, comparing the two.
- Finally, you will evaluate how well you have met a collection of related standards in the **Self-Reflection**.

C Implementation:

- Your programs will build the dictionary by reading data from a file. They will **insert each address into the dictionary patricia tree Stage 2**.
- Your programs will **handle the search for keys**. There are three situations that your programs must handle:
  - Exact matches: output **all** records that match the key (Stage 1 and 2).
  - Similar matches: if there are no exact matches, find **the most similar** key and output its associated records (**Stage 2 only**).
  - No matches being found: if neither exact nor similar matches are found, indicate that there is no match or recommendation (Stage 1 and 2).
- Your programs should also be able to **populate and query** the dictionary with the Assignment 1 linked list approach and the patricia tree approach.
- Your programs should also provide enough information to compare the **efficiency** of the linked list with the patricia tree with an operation-based measure that ignores less important run-time factors (e.g. comparison count).

It is recommended to use your own solution of Assignment 1, and extend it for Assignment 2. But, we will also provide a solution for Assignment 1 if you prefer to work with our code.